Dajin Lee

Ph.D. Candidate, Department of Computer Science and Engineering
Pohang University of Science and Technology (POSTECH), Republic of Korea

dj1122@postech.ac.kr • leerubi.github.io • Google Scholar • LinkedIn

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Research Interests

I investigate how people perceive and act on sensory feedback, and use these insights to design haptic interfaces that shape their experiences and interactions. My work connects perceptual modeling with computational rendering and interactive system design.

- **Multisensory Perception and Modeling:** Investigating and modeling how haptic, visual, and auditory cues interact to shape perceived properties, spatial judgments, and sensory illusions.
- **Perceptually Optimized Haptic Rendering:** Optimizing haptic rendering and actuator placement that account for human perception, actuator constraints, body sites, and contact conditions to produce intended perception and effects.
- **Touch for Media and Exploration:** Designing haptic interactions that enrich media experiences and support exploration, learning, and decision-making in mobile, tangible, and XR systems.

Education

2021.03 – 2027.02 (Expected)
Ph.D. in Computer Science and Engineering
Pohang University of Science and Technology (POSTECH), Republic of Korea
Interaction Laboratory • Advisor: Prof. Seungmoon Choi
Dissertation: *Perceptual Modeling and Optimization of Vibrotactile and Multisensory Feedback: From Spatial Perception to Exploratory Decision-Making*

2020.02
B.S. in Industrial Management and Engineering
Pohang University of Science and Technology (POSTECH), Republic of Korea

Grants and Fellowships

2026
Korea–Canada Research Collaboration Program
National Research Foundation of Korea (NRF)
Principal investigator; independently secured external funding for a research visit to the University of Waterloo.

Selected Honors and Awards

2026 **Special Recognition for Outstanding Review**, ACM SIGCHI
2026 **First Prize**, POSTECH Metaverse Contents Contest
2025 **Honorable Mention**, ICT Challenge, IITP
2024 **Best Paper Award**, Korea Haptics Conference (KHC)
2024 **Best Paper Award Finalist**, IEEE Haptics Symposium
2023 **Honorable Mention**, Hyundai Student Innovation Challenge, KHC

Publications

Journal Articles

[J6] *Springer Virtual Reality* — **Under review**.

[J5] *IEEE Transactions on Haptics (ToH)* — **Under review**.

[J4] **Dajin Lee**, Daehyeon Nam, and Seungmoon Choi. "Augmenting Outdated Museum Exhibits with Embodied and Tangible Interactions for Prolonged Use and Learning Enhancement." *International Journal of Human-Computer Studies (IJHCS)*, 2025.
[Paper] [Demo Video]

[J3] **Dajin Lee** and Seungmoon Choi. "Perceptual Alignment of Spatial Auditory and Tactile Stimuli for Effective Directional Cueing." *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025. Also presented at IEEE VR 2025.
[Paper] [Presentation Video]

[J2] **Dajin Lee**, Daehyeon Nam, Jinhyuk Yoon, Dukchan Yoon, Seokwon Jeong, Keehoon Kim, and Seungmoon Choi. "A Comparative Study of Physical and Haptic Exhibits in an Informal Learning Environment." *IEEE Transactions on Haptics (ToH)*, 2024. Also presented at IEEE World Haptics Conference 2025.
[Paper] [Demo Video]

[J1] **Dajin Lee**, Gyeore Yun, and Seungmoon Choi. "Effects of Contact Force on Vibrotactile Perceived Intensity Across the Upper Body." *IEEE Transactions on Haptics (ToH)*, 2024. Also presented at IEEE Haptics Symposium 2024; **Best Paper Award Finalist**.
[Paper] [Slides]

Conference Papers

[C9] *ACM CHI 2027* — **Under review**.

[C8] *ACM CHI 2027* — **Under review**.

[C7] Duhong Kweon, **Dajin Lee**, Seulbae Kim, and Seungmoon Choi. "HAPTCHA: Eyes-Free Haptic CAPTCHA in Mobile Environments: Design Principle, Security Evaluation, and User Experience" *ACM Symposium on User Interface Software and Technology (UIST)*, 2026.
[Paper] [Supplementary Video]

[C6] Ayush Bhardwaj, Ashish Pratap, Abbas Khawaja, Yapeng Tian, Uison Ju, **Dajin Lee**, Seungmoon Choi, and Jin Ryong Kim. "Touch with Meaning: A Contextual Analysis of Social Touch." *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2026.
[Paper] [Supplementary Video]

[C5] **Dajin Lee**, Soyeon Nam, Minseong Noh, and Seungmoon Choi. "Effects of Frequency and Magnitude on Steering Wheel Vibration Discomfort: A Hierarchical Modeling Approach to Individual Sensitivity." *IEEE Haptics Symposium*, 2026.
[Paper]

[C4] Hyunuk Kim, **Dajin Lee**, and Seungmoon Choi. "Enhancing Body-Penetrating Phantom Sensations Through Multisensory Integration of Sound and Vibration." *IEEE World Haptics Conference (WHC)*, 2025.
[Paper] [Supplementary Video]

[C3] Andreia Valente, **Dajin Lee**, Seungmoon Choi, Mark Billinghurst, and Augusto Esteves. "Modulating Heart Activity and Task Performance Using Haptic Heartbeat Feedback: A Study Across Four Body Placements." *ACM Symposium on User Interface Software and Technology (UIST)*, 2024.
[Paper] [Presentation Video]

[C2] **Dajin Lee**, Jaejun Park, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi. "Participatory Design for In-Vehicle Vibrotactile Warnings on the Driver's Seat." *EuroHaptics*, 2024.
[Paper] [Presentation Video] [Demo Video]

[C1] **Dajin Lee**, Seungjae Oh, Seungmoon Choi, and Bum-Jae You. "Vibrotactile Metaphor of Physical Interaction Using Body-Penetrating Phantom Sensations: Stepping on a Virtual Object." *IEEE World Haptics Conference (WHC)*, 2021.
[Paper] [Presentation Video]

Posters, Extended Abstracts, and Demonstrations

[P2] **Dajin Lee** and Seungmoon Choi. "Spatial Tactile Optimizer: Toward Perceptual Optimization for Multi-Point Vibrotactile Cues on the Torso." *ACM CHI*, 2026 (Extended Abstract).
[Paper]

[P1] **Dajin Lee**, Daehyeon Nam, Seokwon Jeong, Keehoon Kim, and Seungmoon Choi. "A Comparative Study of Physical and Haptic Exhibits in an Informal Learning Environment." *IEEE World Haptics Conference (WHC)*, 2023 (Work in Progress).

[D2] Hyunuk Kim, **Dajin Lee**, and Seungmoon Choi. "Body-Penetrating Tactile Phantom Sensations Enhanced by Sound." *IEEE World Haptics Conference (WHC)*, 2025 (Demonstration).
[Paper] [Supplementary Video]

[D1] **Dajin Lee**, Jaejun Park, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi. "Participatory Design for In-Vehicle Vibrotactile Warnings on the Driver's Seat." *EuroHaptics*, 2024 (Demonstration).
[Paper] [Demo Video]

Selected Research Experience

2026.06 – Present **Visiting Graduate Researcher**
Haptic Experience (HX) Laboratory, University of Waterloo, Canada
Advisor: Prof. Oliver Schneider
Research on multimodal music previews combining audio, text, and vibration [C9], and adaptive haptic shared control for robotic grasping that adjusts assistance based on user resistance to guidance.

2025.07 – 2025.08 **Graduate Research Intern**
Multimodal Interaction (MI) Laboratory, University of Texas at Dallas, USA
Advisor: Prof. Jin Ryong Kim
Contributed to the development of a taxonomy of social touch [C6].

2020.09 – 2020.12 **Research Intern**
Medical Assistant Robotics and Cognitive Haptics (MARCH) Laboratory, POSTECH
Advisor: Prof. Keehoon Kim
Worked on EMG-based control of robotic prostheses.

Invited Talks and Tutorials

2025.07 **Incorporating Qualitative Analysis in Haptics Research** (with Seungmoon Choi)
IEEE World Haptics Conference (WHC), Tutorial
[Course Page] [Slides]

Teaching Experience

2024.09 – 2024.12 **Teaching Assistant**, Introduction to Haptics
POSTECH, Instructor: Prof. Seungmoon Choi

2019.09 – 2021.09 **Teaching Assistant**, Introduction to Statistics for Data Science
POSTECH & POSCO Group University

Mentoring

2024 **Undergraduate Research Mentor**, POSTECH
Mentored one undergraduate student on a research project that led to an IJHCS 2025 publication [J4]. The project also won First Prize in the POSTECH Metaverse Content Contest in 2025.

2025 – 2026 **Undergraduate Research Mentor**, POSTECH
Mentored one undergraduate student from research question formulation through acceptance of their first-author paper at ACM UIST 2026 [C7]. The student also received the Best Undergraduate Research Award at POSTECH in 2025 for this project.

Academic Service

Journal Reviewer

2025, 2026	IEEE Transactions on Haptics (ToH)
2026	International Journal of Human–Computer Studies (IJHCS)
2026	Current Psychology
2026	Frontiers in Psychology
2024	IEEE Transactions on Human-Machine Systems (THMS)

Conference Reviewer

2026, 2027	ACM Conference on Human Factors in Computing Systems (CHI)
2027	IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)
2027	IEEE International Conference on Robotics and Automation (ICRA)
2026	ACM Symposium on Virtual Reality Software and Technology (VRST)
2026	IEEE International Conference on eXtended Reality (ICXR)
2026	AsiaHaptics
2026	IEEE Haptics Symposium
2025	IEEE World Haptics Conference (WHC)
2024, 2026	EuroHaptics

Skills

Methods	**Perceptual Modeling**: Psychophysics, sensory threshold estimation, crossmodal perception. **Optimization**: Constrained optimization, real-time rendering algorithms. **Statistics & Evaluation**: Qualitative and mixed methods, Bayesian hierarchical modeling, mixed-effects regression.
Sensing	**Biosignal Acquisition**: ECG, EMG, EMG-based control. **Motion & Pressure Sensing**: Inertial measurement units (IMUs), pressure sensors, RGB-D motion capture (Microsoft Kinect). **Data Acquisition**: NI DAQmx, sensor integration.
Haptics & Systems	**Haptic Actuation & Rendering**: Vibrotactile, impact, and thermal feedback; kinesthetic force-feedback rendering (Touch/PHANToM devices). **XR Development**: VR and AR implementation, real-time integration of sensing and haptic feedback. **System Evaluation**: In-the-wild mobile deployments, longitudinal user studies.
Tools & Code	Python, MATLAB, R, C/C++, C#, Swift, Unity, NI DAQmx
Languages	Korean (native), English (professional working proficiency)

References

Prof. Seungmoon Choi (Doctoral Advisor)
Professor, Computer Science and Engineering, POSTECH
choism@postech.ac.kr

Prof. Oliver Schneider (Visiting Research Host)
Associate Professor, Management Sciences and Engineering, University of Waterloo
oliver.schneider@uwaterloo.ca

Prof. Jin Ryong Kim (Research Collaborator)
Professor, Computer Science, University of Texas at Dallas
jinryongkim@gmail.com